

Software Process Improvement in Agile Software Development: A Systematic Literature Review

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Abstract—It is recognized the relevance and importance that software process improvement (SPI) and agile development have gained in the field of software engineering. Both are approaches that increase the efficiency and effectiveness of a software development organization and to enhance software products. This paper aims to identify and characterize SPI in agile environments through a systematic literature review, including 423 papers published between 2001 and March/2013. The selected papers were classified according to SPI aspects. Distinct approaches to Agile SPI were identified, the most common is to adapt traditional approaches of SPI in agile environments, but there are novel approaches such as improving people behavior or even improving practices being innovative in SPI consideration. Conducting SPI initiatives in agile environments is quite different from the traditional one, becoming necessary to adapt existing SPI approaches or creating new methods. The main discussion about agile SPI is: what the role of SPI in a context with people and interactions are more valued than process and tools?

Keywords—Software Process Improvement; Agile Software Development; Systematic Literature Review.

I. INTRODUCTION

Software process has gained major attention of the community of software engineering in the last three decades [1,2]. Software processes are human-centred activities and, as such, prone to unexpected or undesired performance and behaviours [3]. It is generally accepted that software processes need to be continuously assessed and improved in order to fulfil the requirements of the customers and stakeholders of the organization [3].

Software Process Improvement (SPI) encompasses the assessment and improvement of the processes and practices involving software development [4]. SPI initiatives are, henceforth, referred to activities aimed to improve the software development process [5]. In recent decades, SPI has been widely studied in the context of “traditional” software development, and its strengths and weaknesses have been largely discussed [6].

Another approach, which adoption is increasingly by software organizations, is the agile software development. This approach is based on velocity of respond to changes, providing competitive advantage. Agile processes should be light of artefacts and activities non-related to coding [7]. Considering the agile mindset, new challenges and opportunities for conducting SPI are also emerging [6].

In this light, this paper provides a systematic literature review on Software Process Improvement in agile environments showing which subjects were studied and also raised from empirical data of primary studies. This paper is organized as follows. Background and related work is presented in Section 2 and our research methodology is presented in Section 3. In Section 4 we describe the results and answer our research questions. We present our conclusions in Section 5.

II. BACKGROUND

A. Software Process Improvement on Agile Development

From the mid-1990s onwards, agile software development principles and methodologies have been increasingly challenging the traditional view of software development and how SPI should be conducted. The main critiques to traditional SPI initiatives are the need of heavy bureaucracy and the innovation constriction [6].

The traditional approach of software development emphasizes up-front contract negotiations where requirements, cost and schedule are fixed and the final product will be delivered at the end of the project lifecycle. Thus, extensive documentation and quantitative monitoring of the product development process plays a central role in traditional approaches of SPI [6].

It has long been recognized the desire of customization and flexibility in working methods, as well as customer involvement and user participation. Agile methodologies are created to address some of these issues. Many of practitioners, method engineers, software engineers and process engineers want to create their own knowledge-oriented methods; documenting their own environment tools and design their software development processes [8].

The continuous improvement in order to find the most suitable software process may not work well in environments susceptible to changes. In many cases, every individual software organization must develop its own practices of continuous self-improvement and establish an appropriate improvement infrastructure [9].

It is argued that the SPI activities should be agile rather than following deterministic and normative based approaches. An agile approach to SPI would be flexible and responsive according local needs, encourage innovation, build projects around those who are motivated, encourage self-organizing

competent teams, and promote sustainable development of the processes [7].

Traditional SPI methods often emphasize the continuous improvement of organizational software development processes, while the principles of agile software development focus on iterative adaptation and improvement of the activities of individual software development teams to increase effectiveness [6].

Agile software development requires a highly untraditional approach to SPI, in which the process improvement knowledge of software developers and teams are acknowledged and valued. The focus of improvement is not in the processes of the company but in individual behaviour within individual teams [6].

B. Related Work

Unterlalmsteiner et al. presented a systematic literature review aimed to identify and characterize evaluation strategies and measurements used to assess the impact of different SPI initiatives. The found seven distinct evaluation strategies, whereas the most measured attribute was Quality (62%), followed by Cost (41%) and Schedule (18%). They conclude that evaluation validity of SPI initiatives is challenged by the scarce consideration of potential confounding factors and the inaccurate descriptions of the evaluation context. Measurements to assess short and mid-term impact of SPI initiatives prevail, whereas long-term measurements tend to be less used [5].

Pino et al. conducted a systematic review in order to analyse the existing approaches towards SPI focused on Small and Medium Enterprises (SME) considering case studies carried out in industry. They found that Capability Maturity Model Integration (CMMI) was the most used model in this companies (38%), followed by the Standard CMMI appraisal model for process improvement (SCAPMI) observed in 16% and ISO 9000 presented in 13% of the papers [10].

Pedreira et al. presented a systematic review of software process tailoring, analysing the existing approaches. The results presents that 33% of processes tailoring initiatives are informal, 35% occurs in organization level and 20% occurs in small and medium companies [11].

Dybå, and Dingsøyr conducted a systematic review of empirical studies of agile software development. Few results were found about process improvement, just suggestions about self-improvement and improvement on general quality [12]. Dingsøyr et al. also conducted other systematic review Researches on agile software development and suggested that many aspects, such as, process improvement still needs further research [13].

III. RESEARCH METHODOLOGY

In this section we describe the design and the execution of the SLR. Furthermore, we discuss threats to the validity of this review. Figure 1 presents the twelve steps adopted in this research process. The steps are described in detail in following subsections.

A. Need For Systematic Review

The motivation of this systematic review came after the investigation of Salo's Thesis [6] and observation of the lack of researches about this subject pointed by Dingsøyr et al [13]. In

order to determine if similar work had already been performed, we searched on:

- ACM Digital Library¹;
- Google Scholar²;
- IEEE³;
- CiteseerX⁴;
- Springer⁵;

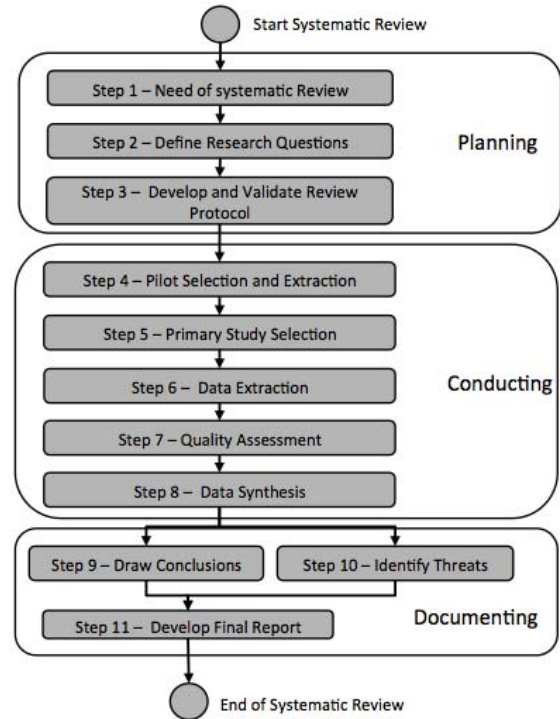


Figure 1. Systematic review steps (based on [15])

We used the following generic search terms looking in the title, keywords, abstracts and full text, when permitted:

Software Process Improvement based on [5] and [10]: "SPI" OR "Software Process Improvement" OR "Process Improvement".

Agile based on [12]: "Agile" OR "Scrum" OR "Extreme Programming" OR "XP" OR "Lean"

SLR based on [5], [14] and [16]: "systematic review" OR "research review" OR "research synthesis" OR "research integration" OR "systematic overview" OR "systematic research synthesis".

Our final generic search string was: Software Process Improvement Terms AND Agile Terms AND SLR Terms. We did not found any systematic review concerned in software process improvement. But, we could not consider starting a new review before elaborate the research questions and verify if other SLRs answer them. Detailed

¹ <http://portal.acm.org>

² <http://scholar.google.com>

³ <http://ieeexplore.ieee.org>

⁴ <http://citeseer.ist.psu.edu>

⁵ <http://link.springer.com/advanced-search>

information can be found on the spreadsheet used to perform this SLR that is available online⁶.

B. Research Questions

To attend our objectives, four research questions were elaborated in order to guide this SLR. They are presented in Table I as following.

C. Creation and Evaluation of a Review Protocol

The reasons to develop a review protocol are: (i) to reduce potential researcher bias, (ii) avoid communication problems among researchers and (iii) permit a replication of the review in the future [15]. Our protocol will be incrementally presented in the following subsections in this Section 3. An experienced researcher in secondary studies in software engineering, which not participate of this work, evaluated our protocol. According his advices and our experiences along the process, we iteratively improved the design of the protocol.

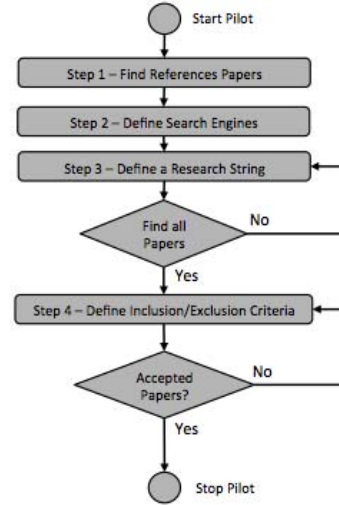


Figure 2. Pilot Test Steps

TABLE I. RESEARCH QUESTIONS

ID	Question	Aim
RQ1	Which approaches are used in Agile-SPI?	Identify which are the general approaches of agile SPI.
RQ2	Which aspects could be extracted from agile SPI Initiatives?	In which categories/aspects of agile SPI could be defined/explained?
RQ3	What are the differences between traditional and agile SPI?	Identify major differences between agile and traditional SPI and how these differences happen on organizations.
RQ4	Could Traditional and agile SPI initiatives be blended?	Are there initiatives to blend agile and traditional SPI?

D. Pilot Selection and Extraction

In order to avoid bias in this SLR, a pilot test was performed aiming to adjust the protocol. The Figure 2 presents the steps of the execution of the pilot. The first step of the pilot was to determine which researches are more important considering SPI in agile environments. In order to discover these papers, we answered to eight specialists about important references about agile SPI. This interview was performed in XP 2011 conference and there were no limitations about the quantity of sources. Table II summarizes the answers of the questionnaires.

The step 2 was related to which search engines we would use. We decided to maintain the same engines used to find related SLR presented in section A.

The step 3 was to define the search string. The first version of the string was based on that one used when seeking other SLRs. That string was adjusted until we find all papers indicated by specialists.

Our final generic string was: ("SPI" OR "Software Process Improvement") AND ("Agile" OR "Scrum" OR "Extreme Programming" OR "XP" OR "Lean" OR "DSDM" OR "FDD")

The step four was to define the inclusion and exclusion criteria. The initial inclusion and exclusion criteria are based on a SLR performed by Beecham et al. [16], and after the adaptation, and some iterations on pilot our final inclusion/exclusion criteria were:

E. Inclusion Criteria

- IC 1 – The research is related to any aspect of SPI and Agile Development;
- IC 2 – The research is a primary study containing empirical results and is written in a scientific format.

F. Exclusion Criteria

- EC1 – The research is a secondary or tertiary study, or the paper presents primary results aiming to teach SPI or agile techniques.
- EC2 – The research is just an abstract or short paper presenting "unevaluated" new idea. Or the paper just publishes lessons learned or experience reports or papers without empirical results. Or the paper presents an expert opinion.
- EC3 – The research is not written in English;
- EC4 – The research is not related to agile SPI or does not answer any of the research questions.

⁶ https://www.dropbox.com/s/i187bai7chxflus/Spreadsheet_SLR_AgileSPI.xlsx?dl=0

TABLE II. ANSWER OF SPECIALISTS ABOUT IMPORTANT RESEARCHES ON SPI

ID	Author(s)	Title	N° of indications
1	Outi Salo [6]	Enabling Software Process Improvement in Agile Software Development Teams and Organizations	8
2	Brede Moe [17]	From Improving Processes to Improving Practice	6
3	Wan, Wang [18]	Empirical Research on Critical Success Factors of Agile Software Process Improvement	5
4	Stamelos, Sfetos [19]	Agile Software Development Quality Assurance	4
5	Salo, Abrahamsson [20]	Integrating agile software development and software process improvement: a longitudinal case study	4
6	Outi Salo [21]	Improving software process in agile software development projects: results from two XP case studies	3
7	Siakas, Elis, Eleni [22]	Agile Methodologies and Software Process Improvement.	1

The goal of our pilot was to find all papers on search and include them after criteria evaluation. The search string was a little different from the original, and the criteria were not modified.

G. Primary Study Selection

With the search string and criteria evaluated, we started the selection of the primary studies. The selection occurred in two phases: (i) select papers based on title, abstract or introduction and (ii) in selected papers accepted of the first phase, applying the inclusion/exclusion criteria considering all the document. Figure 3 presents a summary of our process selection.

We found 8986 papers, where 1381 were duplicated. We did not download nine papers that we need further details (introduction) to take a better decision. And 7452 were discarded in the first step remaining 144 papers to second step. In second step, we analysed the 144 papers and applied the inclusion and exclusion criteria based on all text, and only 32 were selected. The list of 31 selected papers is presented on appendix A of this document.

H. Data Extraction

In this step two researchers extracted the data of all 31 papers. Our first concern was to identify categories of agile SPI subjects. All categories and related research questions are listed in Table III. These categories were not chosen a-priori, they emerged along the extraction and the results related of each category will be presented in Section 4. These categories also answer our RQ2.

I. Quality Assessment

The study quality assessment (Step 7) can be used to guide the interpretation of the synthesis findings and to determine the strength of the elaborated inferences [14]. However, as also experienced by Unterkalmsteiner [5], we found it difficult to assess to which extent the authors of the studies were actually able to address validity threats. Indeed, the quality assessment

we have performed is a judgment of reporting rather than study quality.

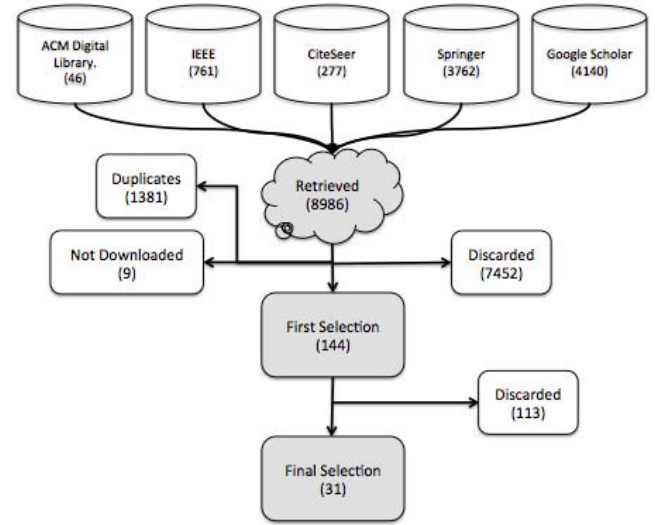


Figure 3 – Selection of papers.

We used the same quality criteria adopted by Dyba and Dingsoir [12], but we did not follow the same process. While Dyba and Dingsoir used quality criteria to exclude studies, we just evaluated overall aspects of the research. Dyba and Dingsoir pointed eleven criteria and just observe if the aspect is present (yes) or absent (no). The summary of results of the quality criteria evaluation is presented in Table IV.

TABLE III. AGILE – SPI CATEGORIES

Category	RQ. Addressed
SPI Approaches	RQ1 / RQ3
SPI Models / Initiatives	RQ1 / RQ3
SPI Related Models	RQ3
Process Assessments	RQ1/ RQ3 / RQ4
Process Tailoring	RQ1 / RQ3
Process Deploying	RQ1 / RQ3
Process Measurement	RQ1 / RQ3
Agile + Traditional SPI	RQ4

J. Threats to Validity

We decided to “jump” to the step 10, because the results of Synthesis (Step 9) and Conclusions (Step 10) will be presented in the next Sections. The threats that we find were:

K. Results Bias

Result bias refers to the general practice of research in publish positive outcomes more than negative ones, when negatives are published [14]. We hope this threat have few impact on the results due our research questions being more related to discover aspects of agile SPI and not measure

performance aspects. We did not restrict the sources but we have to balance the great number of literature sources and the quality of the sources in the intention of gather reliable information. Therefore we decided not to include “grey literature” (technical reports, work in progress, unpublished or not peer-reviewed publications) [14].

TABLE IV. QUALITY CRITERIA

Quality Criteria	Yes	No
1. Is the paper based on research (or is it merely a “lessons learned” report based on expert opinion)?	31 (100%)	0 (0%)
2. Is there a clear statement of the aims of the research?	30 (96,77%)	1 (3,23%)
3. Is there an adequate description of the context in which the research was carried out?	22 (70,96%)	9 (29,04%)
4. Was the research design appropriate to address the aims of the research?	23 (74,19%)	8 (25,81%)
5. Was the recruitment strategy appropriate to the aims of the research?	8 (25,81%)	23 (74,19%)
6. Was there a control group with which to compare treatments?	0 (0%)	31 (100%)
7. Was the data collected in a way that addressed the research issue?	20 (64,51%)	11 (35,49%)
8. Was the data analysis sufficiently rigorous?	15 (48,38%)	16 (51,24%)
9. Has the relationship between researcher and participants been considered to an adequate degree?	11 (35,49%)	20 (64,51%)
10. Is there a clear statement of findings?	31 (100%)	0 (0%)
11. Is the study of value for research or practice?	29 (93,54%)	2 (6,46%)
General Results	64,51%	35,49%

L. Threats to the identification of primary studies

Since it is impossible to know all existing relevant papers about agile SPI we tried to identify some relevant works with experts and conduct a pilot to retrieve all indicated papers. But experts may fail and, our string is certainly incomplete, the terminology (search terms) could be slightly different in good papers may have biased the identification of primary studies regards our efforts to minimize it.

M. Threats to selection and data extraction

One of the main aims of defining a review protocol is to reduce researcher bias [14], we defined objective criteria to include and exclude papers, used pair reviews, used third parties to “untie” conflicted papers, cross-check outcomes and avoid manual searches and snowballing process, but, this action just reduce bias due individual interpretations.

Due the large amount of initially identified studies (8986) we found this strategy impossible to implement within the given timebox. We piloted the paper selection addressing two issues: (i) the selection criteria and the data extraction form were tested for appropriateness, (ii) the agreement between researchers could be assessed and discrepancies streamlined. But, even with piloting, many papers were discarded, in Google Scholar, for example, just 0,3% of the papers were selected as primary study suggesting a problem or with the search string or with the criteria.

IV. RESULTS

A total of 31 studies discuss agile SPI. Before presenting the results for each research question we provide the demographics data of the studies.

A. Overview of the studies

The reviewed papers were published between 2002 and 2012. Our First perceptions about research in agile SPI were the 3 directions followed along the years. In the first stage (2002-2005 – 9 papers) the foundations of agile SPI were still being proposed. Questions like: what is agile SPI? How agile was related with formal initiatives of SPI? How to measure and assess agility? And initial propositions of agile SPI methods were raised at that moment.

The second stage comprehends the period between 2006-2009 (14 papers) and it is highlighted by the experimentation of the “new ideas” in real environments. This stage also the moment where the 3 approaches to agile SPI already emerged, further details of these approaches will be presented in next subsection (RQ1).

In the third stage (2010-2012 – 8 Papers) we found papers concerned in investigate SPI aspects such as tailoring, deploying and measurement. We also find papers presenting results related to specific practices of agile SPI combined to other methods such as lean and CMMI.

Considering research methods, case studies (28, 90%) and industry reports (9, 29,7%) constitute a clear majority of the studies. Also interesting to observe is the lack of an adequate description of the applied research methodology prevented a categorization (9, 22%). Other interesting data is about the confidentiality of industry reports which represents 66% (6 papers) of all industry reports.

RQ1 - Which approaches are used in Agile-SPI?

Salo [6] identified two major approaches for SPI which are: (i) Top-down: which primary goal is related to organizational procedures for improving the organisational software processes. The knowledge transfer is explicit and the external knowledge capture and inert knowledge transfer to support a learning paradigm and the basis for improvement are organisational goals and measurements.

(ii) Bottom up: the primary goal is adapting the process to the contextual needs of individual project teams and improving the effectiveness of individual project teams. Based on Self-organisation of teams using face-to-face communication to establish and update project knowledge in the participants’ heads rather than in documents. And the basis for improvement are based on contextual needs, experience and learning of software developers and regular team reflections.

Primary studies presented both approaches of agile SPI. Considering top-down approach, the majority are related to blending agile and CMMI in process improvement initiatives. All of papers present evidences of how agile can satisfy CMMI requirements (even if this not represents an improvement in process). Other models are proposed by Patel and Rumachandran [P0031] and their agile maturity model and Pikkarainen [P0050] that proposes an agile-CMMI SPI approach to an automotive industry.

Considering bottom-up approaches there are two mindset to conduct agile-SPI. (i) process improvement based on

“improve” team behaviour and (ii) process improvement based on practice improvement. The researches on improve team behaviour started in 2004 with Salo [P0176]. The central idea is to improve the process by improving the team members behaviour to adjust how they should work together. This kind of SPI is also proposed in agile manifesto [23] in one principle **“regular team reflections of software developers in order to become more effective”**.

The other mindset is related to improving practices in order to delivery better software. The central idea is to improve the competence of team members to realize one task in a more effective and efficient way. Borjesson et al. [P0030] proposed the idea in 2006, but Moe's work [P0110] is a rich reference about this kind of agile SPI.

As answer to RQ1, we can suggest three agile SPI approaches. (i) the top-down one; which goal is to “fit” agile practise into any predefined model; (ii) the agile SPI based on improve behaviour; and (iii) the agile SPI based on improve practices.

RQ2 - Which aspects could be extracted from agile SPI Initiatives?

Based on Stamelos and Sfetsos [P0046], Salo [P0093] and Moe [P0110] we present, in Table V, aspects related to agile SPI.

All of these aspects were researched and practiced in some level. Many studies of agile SPI models, standard process and assessments were excluded due to be: only ideas; non-scientific researches; or novel ideas published in short papers. Few papers ([P0031, P0048, P0050 & P0054]) address these issues and clearly needs further research.

The next four aspects are controversial because a considerable quantity of studies was found in our search, but the measurement, tailoring, deploying and knowledge transfer were not related to any kind of improvement. An example is the paper from Conboy & Fitzgerald [P0210], it was excluded from our study because it is not related to SPI.

The last aspect seems to be more defined and, at the same time, divided. If the approach of agile SPI is top-down, the organization is the target. If the approach is bottom-up, the target seems to be the people (and teams), their interactions and capability.

TABLE V. AGILE – SPI CATEGORIES

ID	Aspect
1	Agile SPI models
2	Standard process of agile SPI
3	Assessments of agile SPI
4	Process tailoring
5	Process deploy
6	Process measurement
7	Knowledge transfer
8	Organizational structure to SPI

RQ3 - What are the differences between traditional and agile SPI?

Considering the aspects suggested in Table V, the major difference is the goal of traditional and agile SPI. In traditional SPI, considering top-down or bottom-up approach, the goal is to elaborate a repeatable process that could be improved by lessons learned (maturity). agile SPI the process should be ready for changes, and its improvements means to improve process capability for changes.

Standard processes, modelled by any language or model, which improvements are made based on improvement projects where the process is an asset is few considered in agile process development because, in that way, the process becomes one more artefact to be managed, maintained and mainly, changed.

Agile SPI models should consider flexibility and embraces change, which becomes the next aspect, standard process for agile SPI almost useless. These standard process are only observed when agile is blended with rigorous methods such as CMMI.

Assessment, and measurement, of any agile aspect is all right since that measures being related to teams and the final product not measuring the individuals or the process. Since, considering Bottom-up approaches, we raise one question: How to measure readiness to change? Or how to measure adaptability? What we should measure in order to capture these aspects? The measurements observed in agile process are related to compare what the team produced this iteration with previous one and evaluate what was wrong. Considering top-down approaches, the measurement system tends to move in direction of the model that agile is blended, and consequently, this is not different from traditional (pure) approaches.

Considering tailoring and deploying, again considering bottom-up approaches, it is pointless because if bottom-up approaches are related to improve behaviour or practices there is no standard process to tailor, or model to deploy. When agile teams perform retrospectives it is rare, to have any kind of process or model to deploy or tailor. In top-down approach, again agile SPI go towards the model that agile is blended.

Knowledge transfer in agile SPI is based on communication face-to-face, informal and the learning is based on individual experiences. Traditional approaches consider that knowledge belongs to the organization and its transference to people is based on training and document use. Informal communication can occur, but it is performed in an unstructured way and do not free from bias.

The organizational structure depends on which approach of agile SPI is used. It is top-down, the organization as whole, and the temporary teams of the projects are considered in the SPI scope, and it is similar to traditional approaches of SPI. If bottom-up approach is considered, the people and team are targets of the SPI initiative.

RQ4 - Could Traditional and agile SPI initiatives be blended?

There is no doubt that the question of this answer is yes. But, the real question raised here is: what we want in blending both SPI initiatives? It is contradictory to gain in the same process maturity (by repeating a “repeatable” process) with be ready for changes. How can we reach a high maturity in change our process? Certainly, is not repeating it indefinitely.

Blend agile and CMMI, for example, could help so much traditional approaches to short the iterations size, and we can repeat the process more time, and learning more about it. This acceleration of the learning is good, but it not means as improvement of agility (readiness to change). But, what is the problem in not being agile? Any, we guess but we want to highlight the misuse of the term agility in many of these cases where the objective of SPI initiative is to learn faster and don't be more flexible and adaptable.

V. FINAL CONSIDERATION

Agile SPI is definitely a subject where new opportunities, challenges and perspectives could be addressed due its dual nature of embrace top-down and bottom-up approaches. When toward to top-down approaches the SPI initiative goal tends to be close that one proposed by traditional SPI methods.

When considering bottom-up approaches we can view, in fact, the real challenges of how to get better teams and better products. The two bottom-up perspectives of agile SPI, improve behaviour of teams and improve practices, lays their foundations on individual and teams respecting the first value of agile manifesto [23]: "Individual and Interaction over Process and Tools" representing more suitable options to agile teams interested in get better.

One question that still remains is: What is the purpose of agile SPI? We can infer that teams more ready to changes. But, if the client needs are not related to changes? In traditional SPI approaches the SPI initiatives are related to organizational issues, but, in agile process, the client is too much important and he is invited to work with/into the teams. In agile manifesto [23] we can find the assertive that agile methods were created to provide competitive advantage to the client. Since, which direction agile SPI should follow? Towards the client? Or towards the organization? All these questions raised in the conclusions were not answered by the primary results selected.

The results of this review encourage further research on agile SPI, particularly on the conception of more agile approaches to SPI.

REFERENCES

- [1] D. I. K. Sjöberg, T. Dyba, M. Jørgensen, "The future of empirical methods in software engineering research." Future of Software Engineering (FOSE), Minneapolis, 2007, pp. 358–378.
- [2] N. Wirth, "A brief history of software engineering." IEEE Ann. Hist. Comput., 2008; vol. 30, pp. 32–39.
- [3] A. Fuggetta, "Software process: a roadmap." In Proceedings Conference on The Future of Software Engineering, Limerick, Ireland, 2000, pp. 25–34.
- [4] D. N. Card, "Research directions in software process improvement." in Proceedings 28th Annual International Computer Software and Applications Conference (COMPSAC), Hong Kong, China, 2004, p. 238.
- [5] M. Unterkalmsteiner, T. Gorschek, A.K.M. Islam., R. B. Permadi, R. Feldt, "Evaluation and Measurement of Software Process Improvement - A Systematic Literature Review". IEEE Transactions on Software Engineering, 2011.
- [6] O. Salo, "Enabling Software Process Improvement in Agile Software Development Teams and Organizations." VTT publications, 2007.
- [7] I. Allison, "Towards an Agile Approach to Software Process Improvement: Addressing the Changing Needs of Software Products." In Communications of IIMA. 2005; vol. 30, pp. 67-76.
- [8] K. Siakas, E. Georgidaou, E. Berki, "Agile methodologies and software process improvement." IADIS (International Association for Development of the Information Society). 2005, pp. 412-417.
- [9] A. Birk, H. Rombach. A Practical Approach to Continuous Improvement in Software Engineering. Software Quality. State of the Art in Management, Testing, and Tools, Berlin Springer. 2001.
- [10] F. Pino, F. García, M. Piattini, "Software process improvement in small and medium software enterprises: a systematic review". *Software Quality Journal*, 16(2), 237-261. 2008.
- [11] O. Pedreira, M. Piattini, M. Luaces, R. Brisaboa, "A systematic review of software process tailoring." *ACM SIGSOFT Software Engineering Notes*, 32(3), 1-6. 2007.
- [12] T. Dybå, T. Dingsøyr, "Empirical studies of agile software development: A systematic review". *Information and software technology*, 50(9), 833-859. 2008.
- [13] T. Dingsøyr, P. Nerur, P. Balijepallyd, B. Moe, "A decade of agile methodologies: Towards explaining agile software development." *Journal of Systems and Software* Volume 85, Issue 6, June 2012, Pages 1213–1221. 2012.
- [14] B. Kitchenham, S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Software Engineering Group, Keele University and Department of Computer Science, University of Durham, United Kingdom, Technical Report EBSE-2007-01, 2007.
- [15] J. Biolchini, P. Mian, A. Natali, T. Conte, G. Travassos, "Scientific research ontology to support systematic review in software engineering," *Advanced Engineering Informatics*, vol. 21, no. 2, pp. 133–151, Apr. 2007.
- [16] S. Beecham, N. Baddoo, T. Hall, H. Robinson, H. Sharp. "Motivation in Software Engineering: A systematic literature review." *Information and software technology*, 50(9), 860-878. 2008.
- [17] B. Moe, "From Improving Processes To Improving Practice: Software Process Improvement in Transition from Plan-driven to Change-driven Development". 2011.
- [18] J. Wan, R. Wang. "Empirical Research on Critical Success Factors of Agile Software Process Improvement." *Journal of Software Engineering and Applications* 3(12). 2010.
- [19] G. Stamelos, S. Sfetsos, Agile software development quality assurance. Igi Global, 2007.
- [20] O. Salo, P. Abrahamsson. "Integrating agile software development and software process improvement: a longitudinal case study." *Empirical Software Engineering*, 2005.
- [21] O. Salo, "Improving software process in agile software development projects: results from two XP case studies." *Euromicro Conference*, 2004. Proceedings. 30th. IEEE, 2004.
- [22] K. Siakas, E. Georgiadou, E. Berki. "Agile methodologies and software process improvement." IADIS International Virtual Multi Conference on Computer Science and Information Systems. 26. 2005.
- [23] Agile Manifesto. Available in www.agilemanifesto.org. Last access in Jan. 12. 2015.

APPENDIX A – SELECTED PRIMARY STUDIES

Paper ID	Title	Authors
P0006	A disciplined approach to adopting agile practices: the agile adoption framework.	A. Sidky, J. Arthur, S. Bohne
P0008	A framework to support the evaluation, adoption and improvement of agile methods in practice.	A. Qumer, B. Henderson-Sellers
P0023	Achieving CMMI level 2 with enhanced extreme programming approach	T. Kähkönen, P. Abrahamsson
P0024	Adapting PROFES for use in an agile process: an industry experience report	A. Jedlitschka, D. Hamann, T. Göhlert
P0030	Agile improvement practices in software organizations	A. Börjesson, F. Martinsson
P0031	Agile maturity model (AMM): A Software Process Improvement framework for agile software development practices	C. Patel, M. Ramachandran
P0033	Agile meets CMMI: Culture clash or common cause?	R. Turner, A. Jain
P0040	Agile process improvement: Diagnosis and planning to improve teamwork	M. Ringstad, T. Dingsøyr, N. Moe
P0046	Agile software development quality assurance	I. Stamelos, S. Sftsos.
P0048	Agile SPI: Software Process Agile Improvement, A Colombia approach to software process improvement in small software organizations	J. Hurtado, F. Pino, J. Vidal, C. Pardo
P0050	AHAA-Agile, Hybrid Assessment Method for Automotive, Safety Critical SMEs	M. Pikkarainen, F. McCaffery, I. Richardson
P0051	Am-quick: a measurement-based framework for agile methods customisation	H. Ayed, N. Habra, B. Vanderose
P0054	An approach for assessing suitability of agile solutions: a case study	M. Pikkarainen, U. Passoja
P0055	An approach for using CMMI in agile software development assessments: experiences from three case studies	M. Pikkarainen, A. Mantyniemi
P0057	An evaluation of the degree of agility in six agile methods and its applicability for method engineering	A. Qumer, B. Henderson-Sellers
P0060	An iterative improvement process for agile software development	O. Salo, P. Abrahamsson

P0063	Applying agile and lean practices in a software development project into a CMMI organization	M. Trujillo, H. Oktaba, F. Pino, M. Orozco.
P0093	Enabling software process improvement in agile software development teams and organisations	O. Salo
P0101	Examining the effects of agile methods and process maturity on software product development performance	M. Rönkkö, J. Peltonen, C. Frühwirth
P0110	From Improving Processes To Improving Practice: Software Process Improvement in Transition from Plan-driven to Change-driven Development	N. Moe
P0116	Implementing CMMI using a combination of agile methods	J. Alegria, M. Bastarrica
P0117	Improving agility and discipline of software development with the Scrum and CMMI	K. Łukasiewicz, J. Miler
P0122	Improving software process in agile software development projects: results from two XP case studies	O. Salo
P0125	Integrating agile software development and software process improvement: a longitudinal case study	O. Salo, P. Abrahamsson
P0140	Mapping CMMI level 2 to scrum practices: An experience report	J. Diaz, J. Garbajosa, J. Calvo-Manzano
P0163	Process Increments: An Agile Approach to Software Process Improvement.	A. Abdel-Hamid, M. Abdel-Kader
P0176	Self-adaptability of agile software processes: A case study on post-iteration workshops	O. Salo, K. Kolehmainen, P. Kyllönen, J. Löthman
P0192	SPI agility: How to navigate improvement projects	I. Aaen, A. Börjesson
P0205	Towards an agile approach to Software Process Improvement: addressing the changing needs of software products	I. Allison
P0206	Towards a better understanding of CMMI and agile integration-multiple case study of four companies	M. Pikkarainen
P0224	Using Scrum to guide the execution of software process improvement in small organizations	F. Pino, O. Pedreira, F. García, M. Luaces